Decision report: DEC-20261023-2e7447

GOAL-CZLIFT-1516d5 · RQ-CZLIFT-b9f765 · H-CZLIFT-fba874 · EXP-CZLIFT-1a3453 · review plan REVIEW-CZLIFT-1A3453-EVIDENCE-20261023 · 2026-10-23

Status

This decision takes effect only when the ledger archive TASK-20261023-4d7449 commits and the post-commit verifier accepts it. Until then nothing here is official. This is a toy-tier spectral measurement, not a performance measurement. No benchmark, frontier row or exponent moves.

Decision

REFINE. The run set is valid. It supports the narrow single-character statement below on the 42 power-bearing records, weakly, at preliminary strength and toy tier. It does not discriminate on the hypothesis title. H-CZLIFT-fba874 moves from **approved** to **analyzed**: it is not supported, weakened or rejected. `decision_impact.on_positive` of EXP-CZLIFT-1a3453 is **not** invoked. A successor hypothesis and contract will restate the claim at the instrument's real scope, measure the detection floor, and target the exponent-relevant scaling of Fourier mass with n.

- **Evidence:** EV-CZLIFT-764f54. Direction supports (narrow scope only). **Strength:** preliminary. **Claim tier:** toy ($p \le 571$, 10-bit fields; $n \le 613$; precision 5).
- **Runs:** RUN-CZLIFT-037e5f (seed 20261009) and RUN-CZLIFT-d37ca4 (seed 20261010), both `output_validated` at commit 3d48a8e6 (clean).
- **Review:** VAL-20261022-55676f (TASK-20261022-8ebef8, passed; J1-J3 hold, PTM-1 holds with scope restriction). RT-20261023-dcb217 (TASK-20261023-792040; R1 breaks, R2 breaks, PTM-1 breaks on the recorded scope and holds on the 42 power-bearing records).
- **Earlier records:** approval DEC-20261009-296cee; observation DEC-20261009-6aef7a.
- **Knowledge:** KN-OPEN-d3a6f0 promoted (Fourier mass of torsion-section digit characters; it sharpens KN-OPEN-3417fc).

Scoped claim (verbatim from DEC-20261023-2e7447):

> On the 42 power-bearing records of RUN-CZLIFT-037e5f and RUN-CZLIFT-d37ca4, no single-digit function a_i(x(s_tor([k]P))) or a_i(y(s_tor([k]P))), i = 0..4, has half-range normalised correlation tau >= rho_min(p,n) = sqrt((ln((p-1)n) + 12)/h), h = (n-1)/2, with any single pair (nontrivial additive character of F_p, character of Z/n). The records are 33 CM-canonical class-number-one integer models and 9 random curves with one naive integer lift, 53 <= p <= 571, 47 <= n <= 613, one random generator per record and precision 5. rho_min is 0.28-0.98 (median 0.50), about 1.44x higher for generator-covariant even x-digit leaks at n < 109. No improvement meeting the predefined threshold was observed over these instances, parameters, solver and budget. The 30 records with n <= 41 carry no information. Nothing follows for nonlinear or pseudorandom encodings, multiplicative characters, weaker or sparser leaks, other models, lifts or generators, or larger n.

Observation

Validity was checked first. The plan has 2 trials and both runs exist. The manifests are schema-complete, and the hashes bind the runs to the contract frozen in 172ec3c94f. Every metric was recomputed from the raw rows. An independent implementation replays all 72 records to 8e-14, with all 2160 spike counts equal. The uniform control is clean, so no invalidation rule fires.

- On both seeds every spike count is 0: higher digits, digit 0, shuffled and uniform (36 records × 10 digit rows per seed).
- Closest real approach: −3.942 (RUN-CZLIFT-037e5f, D=-3, p=571, n=541, y2) and −4.070 (RUN-CZLIFT-d37ca4, D=-163, p=347, n=383, x4).
- Blind records: 30 of 72 (all with $n \le 41$) have $h \le T$, so $X \le h \le T$ for any digit sequence. Power-bearing records: 42 ($n \ge 47$; 33 CM, 9 random), with `rho_min` 0.284-0.978 and median 0.502.
- PTM-1 (planted floor(pk/n)) spikes on 42/42 power-bearing records and 0/30 blind ones.
- CHIRP and INV, which determine k, fire on 0/72 records under the frozen rule, and LEGMSB stays ≤ 0.68 T everywhere. Digit 0, which determines k up to sign, is white on 72/72.
- CM D=-3 y-digits show a structural t = 0 elevation from the order-3 automorphism: mean excess +2.60 real and +3.65 shuffled.

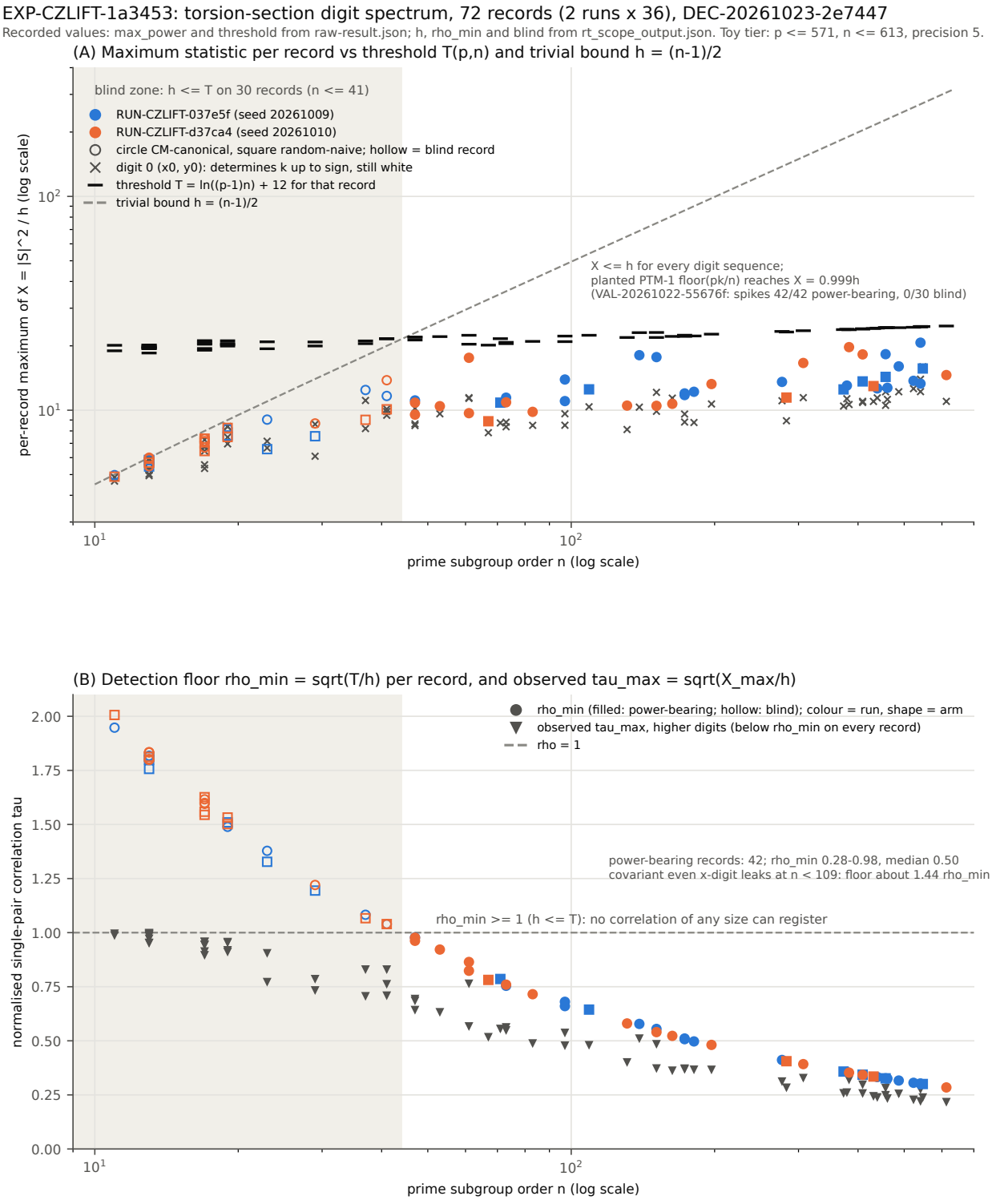


Figure 1: Panel A: the per-record maximum of X over the higher digits (x1-x4, y1-y4; coloured by run, shaped by arm, hollow if blind) and over digit 0 (grey ×), against n. Each record's threshold T(p,n) is drawn as a black tick, and the trivial bound h = (n-1)/2 as a dashed line. The planted PTM-1 leak reaches X = 0.999h. Panel B: the detection floor `rho_min = sqrt(T/h)` per record and the observed `tau_max = sqrt(X_max/h)` of the higher digits. Data: `plot_data.json` (values from both `raw-result.json` files and `rt_scope_output.json`, cross-checked by `plot.py`); regenerate with `plot.py`.

Comparison

comparison	result	grade
frozen criteria	Success met on both seeds; falsification not met; no invalidation rule fires.	as frozen
real vs uniform	Uniform row max = ln N - 0.92 + 0.90 G on 420 rows. Real higher digits outside D=-3 y have mean excess −0.29, against −0.28 for uniform.	randomized

real vs shuffled	The control shares the t = 0 term and samples from the full range, reintroducing the negation mirror; on D=-3 it inherits mu_6 orbits. Per-row spike probability is up to 0.167 against a stated e^-12, and P(zero on both seeds) is about 0.47. The bias runs toward the hypothesis.	not isolated
D=-3 real vs symmetry-matched null	DC mean rank 0.44; non-DC +0.14 above the matched mean; Fisher chi^2 = 76.0 on 78 dof. No hidden excess.	explained
real vs planted PTM-1	Detected on 42/42 power-bearing records: the instrument has power for gross linear leaks.	power shown
CM arm vs random arm	0 spikes on both arms, but 33 vs 9 power-bearing records and one naive lift each.	not isolated

Inference

- The data exclude exactly the single-pair class in the scoped claim. On the pair j = 0 the excluded event is a digit-value bias, not a dependence on k.
- On the title claim ("no p-adic digit of the torsion section correlates with the discrete logarithm") the evidence does not discriminate. Digit 0 carries k and is white, and so is a uniform digit. On that claim the experiment is inconclusive, not supportive.
- The clause "exactly as for the same digits re-indexed by a random permutation" held by chance, under a control biased toward the hypothesis.
- Resource reading. s_tor is a computable homomorphism, so digit characters have exact shift-query access, and their Fourier mass tau is the parameter of a Fourier-learning hidden-number attack (about tau^-2 polylog(n) evaluations). The red team estimates the breakeven against rho at about n^-1/4 up to polylog factors, roughly 2^-53 at 256-bit n. This experiment's floor is tau ≥ 0.28 at n ≤ 613. The measurement is not an obstruction and closes no lane. The exponent-relevant unknown is filed as KN-OPEN-d3a6f0.

Limitation

- Toy tier: p ≤ 571 and n ≤ 613. No transfer to larger n or cryptographic sizes, and no statement about any deployed curve.
- Narrow class: one digit at a time, additive characters, single pairs, the half range, one generator per record and one Weierstrass model per curve. Precision 5 gives digits 0-4.
- Strength is capped at preliminary by the not-isolated shuffled comparison and by the unnumbered random-model null, which fails on D=-3 y-digits, digit 0 and the shuffled series.
- The falsification rule has two readings (per row and per digit aggregate). This is moot for zero counts and load-bearing for any successor.
- The leak-class power figures are single seeded instances from the red team, and the breakeven is an order-of-magnitude estimate.
- This report was composed without a shell. The session renders graph.svg and report.pdf. The subagent harness refused the .md writes, so the session writes analysis.md and KN-OPEN-d3a6f0.md verbatim from the coordinator handback.

Linked reports

- Validation: `reports/validation-20261022/validation_report.yaml` (VAL-20261022-55676f).
- Red team: `reports/redteam-20261023/redteam_report.yaml` (RT-20261023-dcb217).
- Analysis: experiments/EXP-CZLIFT-1a3453/analysis.md.
- Run reports: this experiment has no `reports/run-*` directory. Its run records are under `runs/RUN-CZLIFT-037e5f/` and `runs/RUN-CZLIFT-d37ca4/`.

Canonical graphs checked

knowledge/frontiers/ecdlp/generic-rho/KR-RHO-53c89f.yaml and knowledge/frontiers/ecdlp/index-calculus/KR-IC-cd159a.yaml were checked and left **unchanged**: there is no cost claim and no exponent moves. The only new graph is graph.svg in this folder.

Next actions

- **NA-1 (session, mechanical):** `--check` EV-CZLIFT-764f54, DEC-20261023-2e7447 and KN-OPEN-d3a6f0. Write analysis.md and KN-OPEN-d3a6f0.md from the handback. Run plot.py, which cross-checks every value and writes graph.svg. Compile this file to report.pdf and inspect it. Regenerate knowledge/INDEX.md and run `tools/validate_ledger.py`.
- **NA-2:** Run the ledger archive TASK-20261023-4d7449 alone, staging the exact paths listed in DEC-20261023-2e7447. Verify it, push, and open or refresh a ready PR titled "evidence: EV-CZLIFT-764f54/DEC-20261023-2e7447" that names KN-OPEN-d3a6f0.
- **NA-3:** GOAL-CZLIFT-1516d5 stays active. Checkpoint it with exactly one next action, NA-4 being this lane's proposal.
- **NA-4 (refine successor):** A refined hypothesis on the growth exponent of the maximal full-range Fourier mass of digit characters of s_tor, against n^-1/2 (random) and n^-1/4 (breakeven). The successor contract needs:
  - power-bearing records only, with per-record rho_min published;
  - a symmetry-matched null, with j = 0 removed from the spike count;
  - comparison per row and per pair;
  - planted-leak power curves;
  - generator covariance;
  - n up to about 10^4-10^5;
  - a numbered HEUR null with validation rows at the same scale;
  - a Legendre arm, plus a digit-0 Weil-bound nearby-object control;
  - two or more lifts per random curve.
- **NA-5 (synthesis, RQ-CZLIFT-b9f765 wording):**

  - Qualify “a computable canonical section would be a full break”: `s_tor` exists, is computable, and is an isomorphism, so any gain must come from the encoding.
  - Do not list EXP-CZLIFT-1a3453 as closing the canonical-section clause.
  - Require a detection floor and blind zone for every spectral instrument.
  - Name the coordinate model and the digit encoding of the section.
- **NA-6 (candidate):** Attempt a Weil-type bound for the digit-1 character sum via the Greenberg transform (KN-OPEN-d3a6f0 Q2), going through the KN-TECH-080 gate first.